

Spatial ML - class 11 - Surface modeling methods

Total points 100/100 ?

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1. What is the main goal of Kernel Density Estimation (KDE) in spatial analysis? *10/10

- ☐ To cluster points into groups
- ☒ To convert point data into a continuous surface representing intensity
- ☐ To assign points to administrative regions
- ☐ To calculate distances between points



2. Which R function from *spatstat* is typically used to compute kernel density for point patterns?

*10/10

- ☐ kde()
- ☒ density.ppp()
- ☐ smooth.sf()
- ☐ rasterize()

3. Why is it worth transforming spatial data to a projected CRS (e.g., EPSG:2180) before KDE?

*10/10

- ☐ To reduce file size
- ☐ To allow plotting in ggplot2
- ☒ To ensure distances are measured in meaningful units
- ☐ To simplify attribute tables



4. What is the role of the bandwidth parameter in KDE? *

10/10

- ☐ It determines the number of grid cells
- ☒ It controls the smoothness of the density surface
- ☐ It selects the coordinate system
- ☐ It defines the study area boundary

5. What happens if the bandwidth is set too small? *

10/10

- ☐ The surface becomes overly smooth
- ☐ The surface becomes too generalized
- ☒ The surface shows too many local peaks
- ☐ The computation fails



6. Which object type is required by *spatstat* for KDE computation? *

10/10

- ☐ sf
- ☐ SpatRaster
- ☒ ppp
- ☐ data.frame

7. What does a relative spatial risk surface represent? *

10/10

- ☒ The ratio of two KDE surfaces
- ☐ The distance between clusters
- ☐ The probability of clustering using DBSCAN
- ☐ The absolute number of points in each location



8. When aggregating point data to a grid, what is typically calculated per grid cell? *10/10

- ☐ Mean coordinate values
- ☐ Distance to nearest neighbor
- ☒ Number or density of points within each cell
- ☐ Kernel bandwidth

9. What is one key disadvantage of aggregating point data to polygons * 10/10

- ☐ It increases computational complexity
- ☐ It requires raster data
- ☐ It prevents visualization
- ☒ It may lead to loss of spatial detail and MAUP issues



10. Which of the following best describes the difference between KDE and grid-based aggregation? *10/10

- ☒ KDE produces a continuous surface, grid aggregation produces discrete values per cell
- ☐ KDE uses polygons, grid aggregation uses points
- ☐ KDE requires no parameters, grid aggregation does
- ☐ KDE only works with raster data

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